PHASE 1

Biomedical Fundamentals

ECG Signal Processing & HRV Theory

HRV Feature Extraction & Context-Aware

False Positive Analysis for Atrial Fibrillation Detection

Dataset	:	CACHET-CADB (24 subjects, ~122 hours)
Python	:	3.13.7 | NumPy 2.5.1 | Matplotlib 3.11.0
Phase	:	1 of 18 — Biomedical Fundamentals
Lessons	:	8 lessons | 30 viva questions | 4 mini assignments

CONTENTS

▶ 1.1 Heart Anatomy & Electrical Conduction

▶ 1.2 ECG: What It Is and How It Works

▶ 1.3 ECG Waves: P, QRS, T — Clinical Meaning

▶ 1.4 RR Interval: Foundation of HRV

▶ 1.5 Heart Rate Variability (HRV)

▶ 1.6 Normal Rhythm vs Atrial Fibrillation

▶ 1.7 ECG Noise Types & Motion Artifacts

▶ 1.8 Signal Processing & Filtering Basics

+ ECG Visualizations (6-panel Figure)

+ HRV Statistics Demonstration

+ Sampling Frequency Demo

+ 30 Viva Questions with Answers

LESSON 1.1 & 1.2 — Heart Anatomy & The ECG

Understanding the source of the ECG signal

1.1 HEART ANATOMY

- **Right Atrium (RA)**
  → Receives deoxygenated blood from body
- **Right Ventricle (RV)**
  → Pumps blood to the LUNGS
- **Left Atrium (LA)**
  → Receives oxygenated blood from lungs
- **Left Ventricle (LV)**
  → Pumps blood to the ENTIRE BODY

ELECTRICAL CONDUCTION PATH

SA Node (Right Atrium) → Natural pacemaker, 60-100 BPM

↓

AV Node (Junction) → Delays signal 120-200 ms

↓

Bundle of His → Fast pathway down the septum

↓

Bundle Branches Left + Right → Each ventricle

↓

Purkinje Fibers → Fine network, fires every muscle cell

↓

Ventricles CONTRACT (bottom to top — most efficient)

REAL-WORLD INTUITION:

Normal heart = Stadium wave: ONE person starts,

wave spreads in organised, predictable pattern.

AF = Thousands of people standing randomly

— complete CHAOS, no effective pump action!

1.2 WHAT IS AN ECG?

ECG = Electrocardiogram

"Electro" = Electrical activity

"Cardio" = Heart

"Gram" = Recording (graph)

A graphical recording of the heart's

electrical activity over time.

HOW IT WORKS (step by step):

① Heart fires electrical impulse

② Signal spreads through body (like ripples)

③ Skin electrodes detect tiny voltage (mV)

④ Amplifier boosts the signal

⑤ ADC samples at 256 Hz (256×/second)

⑥ Computer stores: [0.02, 0.15, 1.23, ...]

⑦ Plot Voltage (mV) vs Time (s) = ECG!

SAMPLING FREQUENCY (Critical!)

```
CACHET-CADB uses  fs = 256 Hz

  256 measurements per second

  1 sample every 1/256 = 3.9 ms

  5-min recording = 76,800 data points
```

NYQUIST THEOREM:

```
fs  ≥  2 × f_max
```

ECG max frequency ≈ 40 Hz (QRS)

Minimum required fs = 2 × 40 = 80 Hz

CACHET-CADB: 256 Hz → 6.4× safety ✓

KEY FORMULAS:

```
HR (BPM) = 60,000 / RR (ms)

RR (ms)  = 60,000 / HR (BPM)

Time res = 1000 / fs  ms/sample
```


MINI ASSIGNMENT 1.1 & 1.2

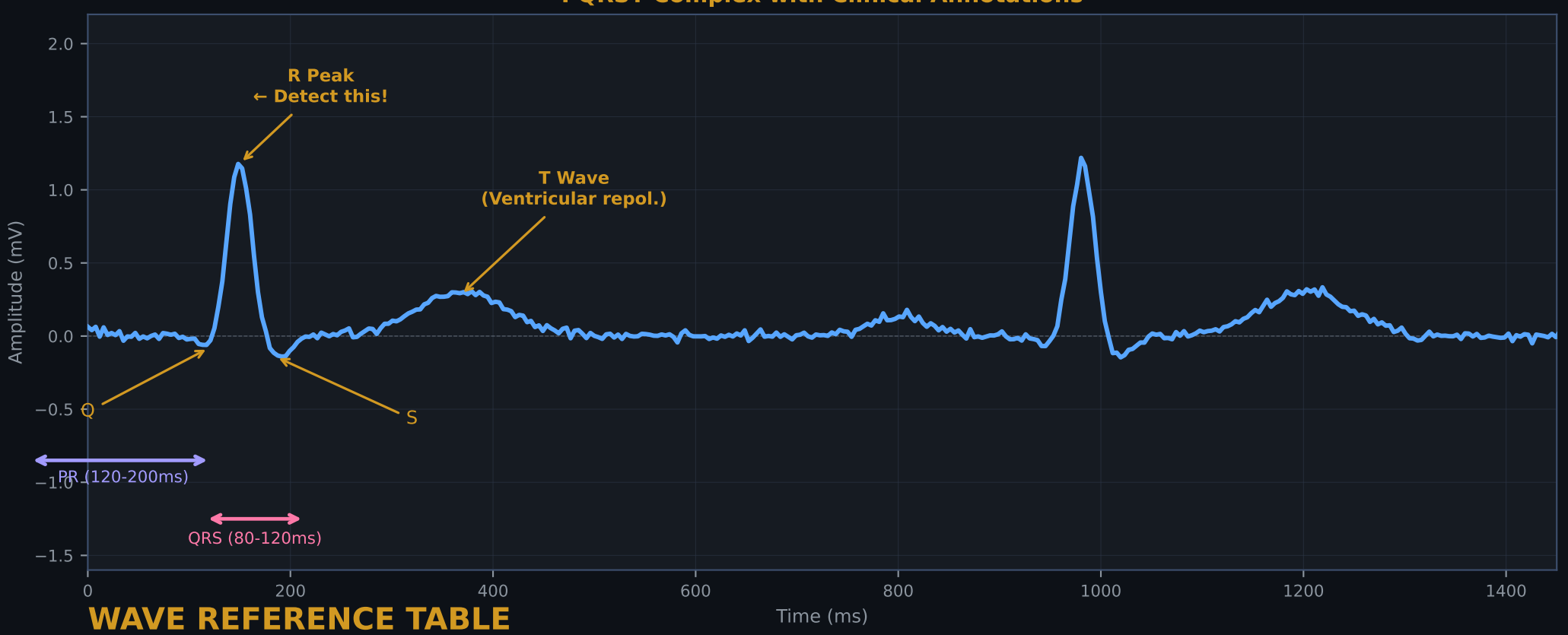
Q1: HR = 85 BPM → RR = ? A: 60,000/85 = 705.9 ms

Q2: RR = 1200 ms → HR = ? A: 60,000/1200 = 50 BPM

Q3: 10-min ECG @ 256 Hz → samples? A: 10×60×256 = 153,600

LESSON 1.3 & 1.4 — ECG Waves & RR Interval

Understanding the PQRST complex and timing measurements



WAVE REFERENCE TABLE

Wave	Represents	Duration	Amplitude	Clinical Significance
P wave	Atrial depolarization	80-120 ms	0.1-0.3 mV	⚠ ABSENT in AF
PR itvl	AV node conduction delay	120-200 ms	—	Prolonged = AV block
Q wave	Septal depolarization	<40 ms	<0.25 mV	Deep Q = old infarct
R wave	Ventricular depol. peak ★ DETECT	~10 ms	0.5-2.0 mV	Our timing reference
S wave	Late ventricular depol.	~20 ms	Variable	—
QRS cplx	Full ventricular contraction	80-120 ms	—	Wide = bundle block
ST seg	Ventricular plateau	80-120 ms	~0 mV	⚠ Elevated = MI!
T wave	Ventricular repolarization	160-320 ms	0.1-0.5 mV	Inverted = ischemia
QT itvl	Depol. + repol. combined	350-440 ms	—	Long QT = arrhythmia risk

1.4 RR INTERVAL

```
RR (ms) = (Sample_R2 - Sample_R1) / fs × 1000

Example:  R1=512, R2=769, fs=256 Hz

  RR = (769-512)/256 × 1000 = 1003.9 ms  ≈  60 BPM
```

VALID RR RANGE:

```
Min RR = 300 ms       → 200 BPM    (Physiological maximum heart rate)

Max RR = 2000 ms      → 30 BPM     (Physiological minimum heart rate)
```

RR PATTERN COMPARISON:

NORMAL: 856 | 862 | 851 | 869 | 845 ms

(±30-50 ms variation)

AF: 612 | 845 | 534 | 923 | 1102 ms

(±300-600 ms variation!)

LESSON 1.5 & 1.6 — HRV & Atrial Fibrillation

What HRV measures and how AF breaks the normal pattern

1.5 HEART RATE VARIABILITY (HRV)

HRV = the variation in time between consecutive heartbeats.

It is NOT heart rate — it is how much the heart rate CHANGES.

HIGH HRV — Healthy ANS, adaptable
→ Athletes, youth, relaxation

LOW HRV — Poor ANS function, rigid
→ Heart failure, diabetes, stress

CHAOTIC HRV — Pathological irregular
→ ATRIAL FIBRILLATION ← Our target

KEY HRV FORMULAS:

```
MeanNN = mean(RR_i)
         # Average RR interval (ms)
SDNN =  std(RR_i) = sqrt[Σ(RR_i - MeanNN)² / (N-1)]
         # Overall variability
RMSSD = sqrt[mean((RR_{i+1} - RR_i)²)]
         # Beat-to-beat variability
pNN50 = count(|RR_{i+1}-RR_i|>50ms) / (N-1) × 100 %
         # High variability fraction
```

ACTUAL CODE OUTPUT (from your script):

```
NORMAL RHYTHM:
  MeanNN : 856.90 ms  →  70.0 BPM
  SDNN   :  10.33 ms  (low = regular)
  RMSSD  :  18.17 ms
  pNN50  :   0.0 %   ✓ Normal

ATRIAL FIBRILLATION:
  MeanNN : 717.80 ms  →  83.6 BPM
  SDNN   : 200.65 ms  ⚠ 19× HIGHER!
  RMSSD  : 373.17 ms  ⚠ 20× HIGHER!
  pNN50  : 100.0  %   ⚠ All beats differ
```

1.6 ATRIAL FIBRILLATION

Feature	Normal Rhythm	Atrial Fibrillation
P waves	Present, regular	⚠ ABSENT (f-waves)
QRS	Regular spacing	⚠ Irregular
RR intervals	Small variation ±5%	⚠ CHAOTIC variation
Heart rate	60-100 BPM	100-180 BPM
AV node	Passes every SA impulse	Randomly passes impulses
SDNN	30-100 ms	⚠ Often >150 ms
pNN50	5-30%	⚠ Often >50%

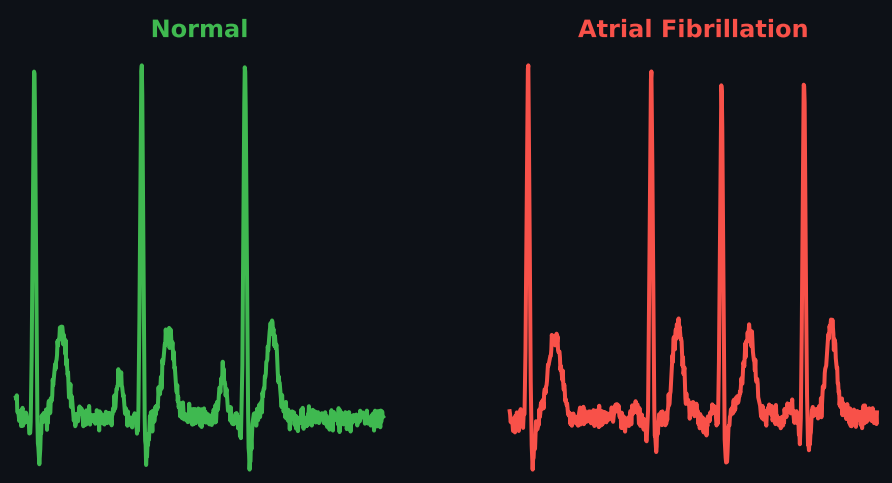
⚠ WHY AF CAUSES STROKES:

Atria quiver → blood pools in

left atrium → clot forms →

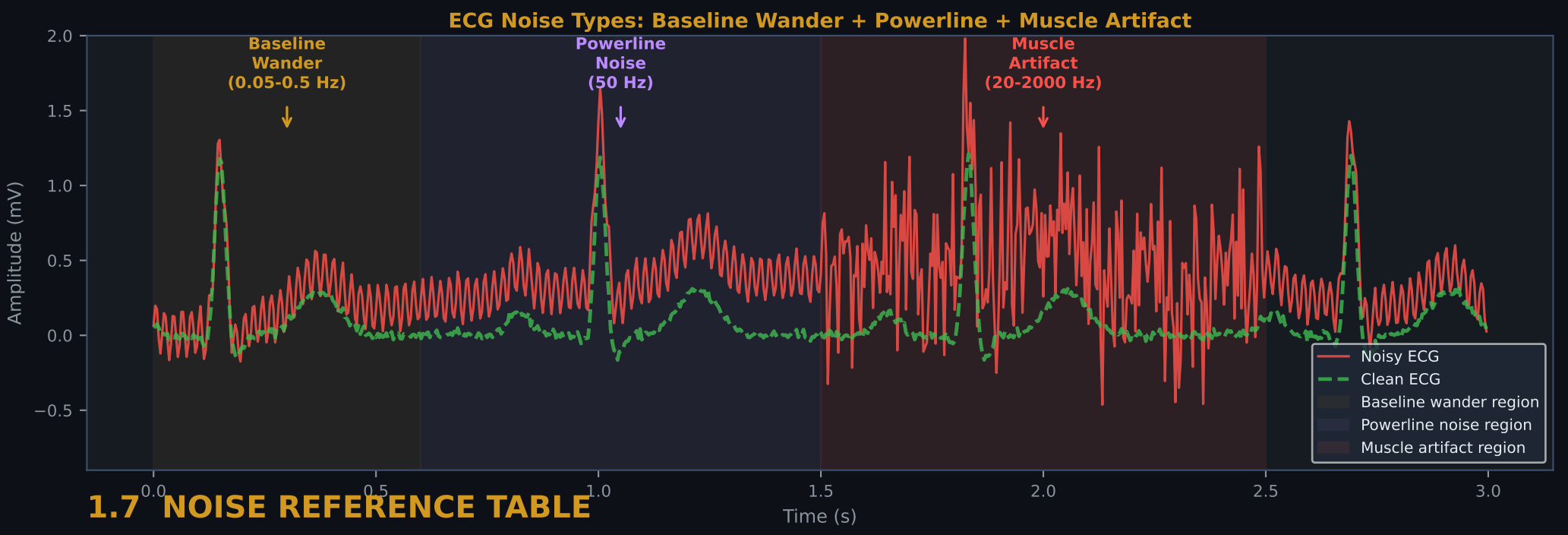
clot travels to brain → STROKE

5× HIGHER stroke risk than normal!



LESSON 1.7 & 1.8 — ECG Noise & Signal Processing

Understanding noise types and how filtering cleans the ECG



1.7 NOISE REFERENCE TABLE

Type	Source	Frequency	Visual	Filter Used
Baseline Wander	Breathing / sweat / movement	0.05-0.5 Hz	Slow drift	High-pass @ 0.5 Hz
Powerline EMI	Electrical outlets (50 Hz India)	50 Hz exactly	Regular buzz	Notch @ 50 Hz
Muscle (EMG)	Skeletal muscle near electrode	20-2000 Hz	Spiky noise	Low-pass @ 40 Hz
Motion Artifact	Electrode moving on skin	0.1-10 Hz	Large jumps	Bandpass filter

THE MOTION → FALSE POSITIVE CHAIN (Our Research!)

Patient walks fast

↓ Electrode moves → motion artifact spikes

↓ R-peak detector: "these look like R peaks!"

↓ Spurious random RR intervals created

↓ SDNN↑↑ RMSSD↑↑ pNN50↑↑ (all AF-like!)

↓ **Classifier predicts: AF ← FALSE POSITIVE!**

→ **CACHET-CADB records accelerometer (MAI)**

We prove: high MAI → more false positives

1.8 SIGNAL PROCESSING BASICS

ECG Frequency Content

0.0 - 0.5 Hz	Baseline wander	**REMOVE**
0.5 - 10.0 Hz	P wave / T wave	**KEEP**
5.0 - 40.0 Hz	QRS complex	**KEEP ★**
50.0 - 50.5 Hz	Powerline noise	**REMOVE**
50.5 - 60.0 Hz	Muscle noise	**REMOVE**

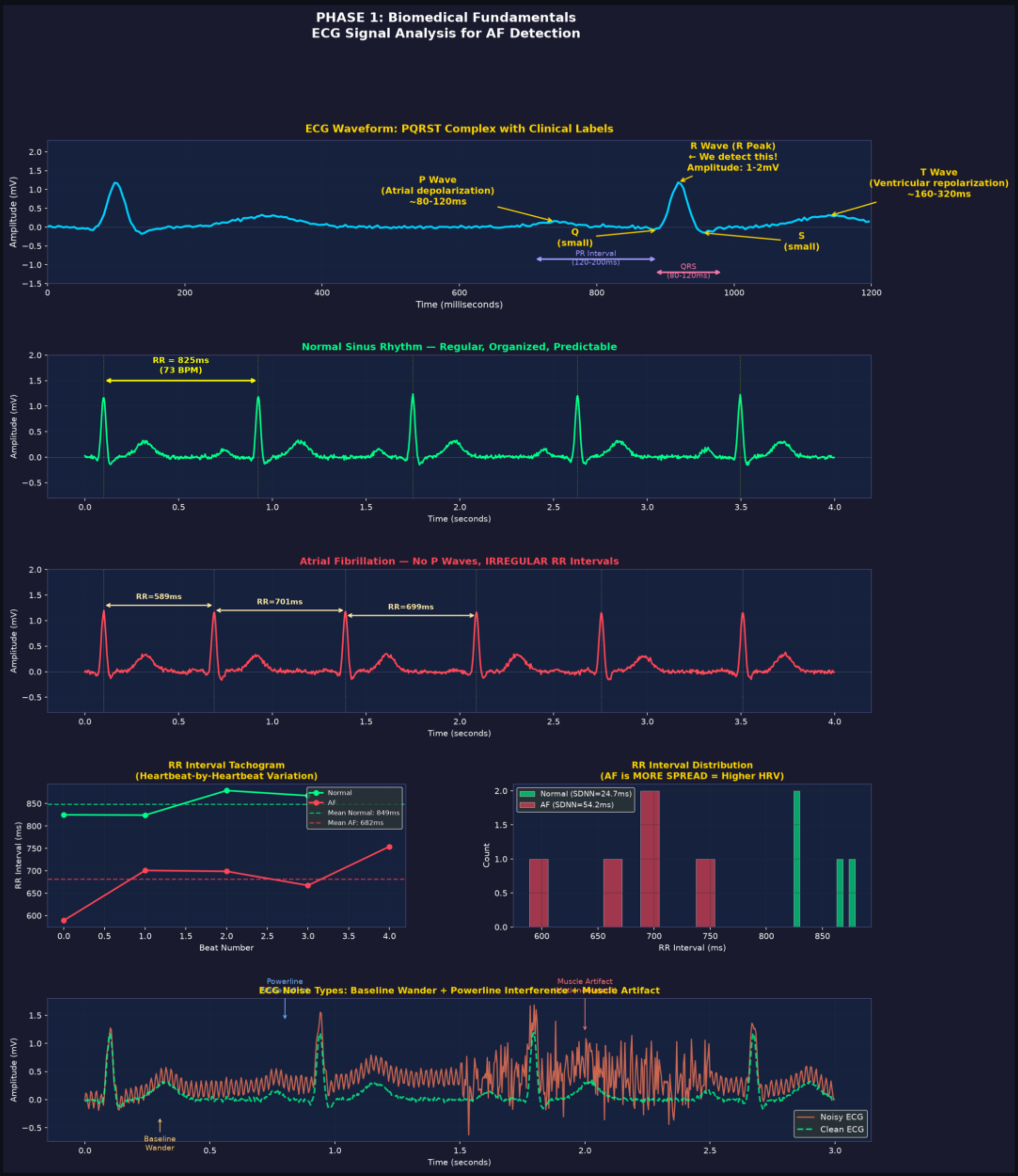
BUTTERWORTH BANDPASS FILTER:

- Keeps frequencies: 0.5 Hz to 40.0 Hz
- Removes: baseline wander + high-freq noise
- Order N = 4 (steeper cutoff than N=2)
- "Maximally flat" → no distortion in passband
- Implementation: scipy.signal.butter() + filtfilt()

Raw ECG → **High-pass 0.5 Hz** → **Low-pass 40 Hz** → **Notch 50 Hz** → **Clean ECG**

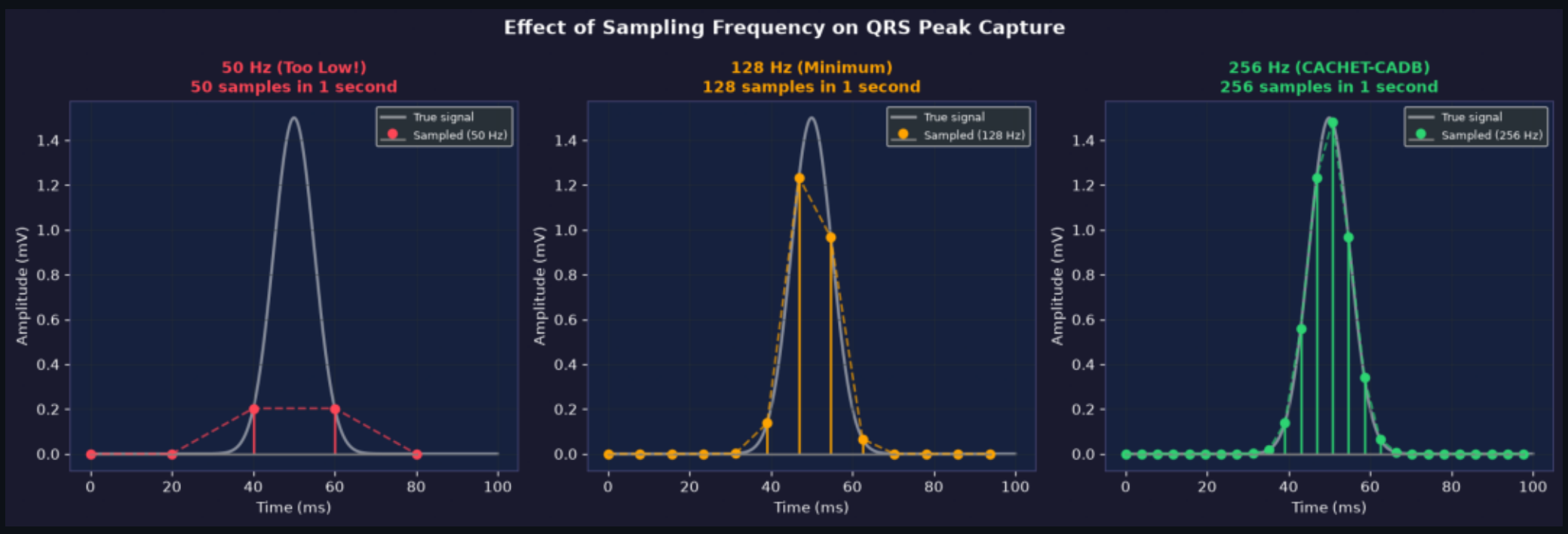
GENERATED FIGURE — ECG Signal Analysis

6-panel educational visualization (output of Phase01_Biomedical_Fundamentals.py)



GENERATED FIGURE — Sampling Frequency Demo

Why CACHET-CADB uses 256 Hz — effect on QRS peak capture



INTERPRETATION:

50 Hz (Red):	Only 5 samples/100ms → completely misses peak height. 7.5× error in amplitude!
128 Hz (Orange):	13 samples/100ms → close but still misses true maximum. Acceptable minimum.
256 Hz (Green):	26 samples/100ms → accurately traces the QRS curve. CACHET-CADB uses this. ✓

At 50 Hz, the R peak might be detected at 0.2 mV instead of 1.5 mV → missed beats or false detections!

NYQUIST APPLIED TO ECG:

```
ECG maximum frequency: ~40 Hz (QRS complex)

Nyquist minimum:  2 × 40 = 80 Hz

CACHET-CADB:      256 Hz  →  6.4× safety margin

Time resolution:  1000/256 = 3.9 ms per sample
```

VIVA QUESTION BANK — Phase 1 (30 Questions)

Prepare these for your final year viva examination

Q1. What does ECG stand for?

A: Electrocardiogram — records the heart's electrical activity over time.

Q2. Name the 4 chambers of the heart.

A: Right Atrium, Right Ventricle, Left Atrium, Left Ventricle.

Q3. What is the SA node?

A: Sinoatrial node — the natural pacemaker in the right atrium; fires at 60-100 BPM.

Q4. Why does the AV node delay the signal?

A: To allow the atria to finish contracting before the ventricles start.

Q5. What does the P wave represent?

A: Atrial depolarization — the electrical event causing atria to contract.

Q6. What does the QRS complex represent?

A: Ventricular depolarization — the main pumping contraction of the heart.

Q7. What does the T wave represent?

A: Ventricular repolarization — the heart muscle recovering and resetting.

Q8. Why is the R wave used as the timing reference?

A: It is the tallest and sharpest feature → highest signal-to-noise ratio.

Q9. What is the RR interval formula?

A: RR (ms) = (Sample_R2 - Sample_R1) / fs × 1000

Q10. Calculate: R1=512, R2=769, fs=256. What is RR?

A: RR = (769-512)/256 × 1000 = 1003.9 ms ≈ 60 BPM

Q11. What is the valid RR range and why?

A: 300-2000 ms (= 200 BPM to 30 BPM physiological extremes). Outside = artifact.

Q12. What is Heart Rate Variability (HRV)?

A: The variation in time between consecutive heartbeats (NOT the heart rate itself).

Q13. What does SDNN measure?

A: Standard deviation of NN intervals — reflects overall autonomic variability.

Q14. What does RMSSD measure?

A: Root Mean Square of Successive Differences — reflects parasympathetic activity.

Q15. What does pNN50 mean?

A: % of consecutive RR differences >50 ms. High value = high variability.

Q16. What is Atrial Fibrillation (AF)?

A: A cardiac arrhythmia with chaotic atrial firing, absent P waves, and irregular RR intervals.

Q17. Name the 3 ECG hallmarks of AF.

A: 1. Absent P waves (replaced by f-waves) 2. Irregular QRS 3. Chaotic RR intervals

Q18. Why does AF cause strokes?

A: Atria quiver → blood pools in left atrium → clot forms → travels to brain.

Q19. What is the stroke risk increase in AF patients?

A: 5× higher stroke risk compared to normal sinus rhythm.

Q20. What is sampling frequency? State it for CACHET-CADB.

A: Number of samples per second (Hz). CACHET-CADB: 256 Hz.

Q21. State the Nyquist theorem.

A: To capture frequency f_max, sample at fs ≥ 2 × f_max. ECG: 80 Hz min.

Q22. What is aliasing?

A: When sampling rate is too low, high-freq signals appear as false low-freq signals.

Q23. Name 4 ECG noise types and their frequencies.

A: 1. Baseline wander (0.05-0.5 Hz) 2. Powerline (50 Hz) 3. Muscle (20-2000 Hz) 4. Motion (0.1-10 Hz)

Q24. What is a bandpass filter?

A: Keeps frequencies within a range (0.5-40 Hz) and removes all others.

Q25. Why use Butterworth filter for ECG?

A: Maximally flat passband — does not distort the signal it keeps.

Q26. What causes motion-related false positives?

A: Physical movement → electrode motion → artifact peaks → spurious RR → AF-like HRV.

Q27. What is MAI in CACHET-CADB?

A: Movement Acceleration Index — quantifies motion from accelerometer data.

Q28. What is paroxysmal AF?

A: AF that comes and goes unpredictably; not always present when ECG is recorded.

Q29. What is a wide QRS complex (>120ms)?

A: Indicates bundle branch block — abnormal conduction pathway in ventricles.

Q30. What does elevated ST segment indicate?

A: Myocardial infarction (heart attack) — acute ischemia of ventricular muscle.

PHASE 1 SUMMARY — Key Numbers & Completion Checklist

Everything you must know cold for viva and project implementation

CRITICAL PARAMETERS TO MEMORIZE

Parameter	Value	Unit	Reason
Sampling Frequency	256	Hz	CACHET-CADB standard
Time Resolution	3.9	ms	1000/256
Min Valid RR	300	ms	200 BPM max (physiological)
Max Valid RR	2000	ms	30 BPM min (physiological)
Bandpass Low	0.5	Hz	Remove baseline wander
Bandpass High	40.0	Hz	Remove muscle noise
Notch Frequency	50	Hz	India powerline interference
HRV Window	5	min	Clinical Task Force standard
Filter Order	4	—	Butterworth sharpness
Normal SDNN	30-100	ms	Clinical reference range
AF SDNN (typical)	>150	ms	Pathological range
Normal pNN50	5-30	%	Clinical reference
AF pNN50 (typical)	>50	%	Pathological range
Total Subjects	24	—	CACHET-CADB dataset
Python Version	3.13.7	—	This project requirement

FORMULA QUICK-REFERENCE

```
HR (BPM) = 60,000 / RR (ms)

RR (ms) = (R2 - R1) / fs × 1000

Nyquist: fs ≥ 2 × f_max
```

```
SDNN  = sqrt[Σ(RR_i - MeanNN)² / (N-1)]

RMSSD = sqrt[mean((RR_{i+1} - RR_i)²)]

pNN50 = count(|ΔRR|>50ms) / (N-1) × 100
```

PHASE 1 COMPLETION CHECKLIST

- ☐ I can explain what an ECG measures (not just "it measures the heart")
- ☐ I can name ALL PQRST waves and what they represent physiologically
- ☐ I can calculate RR interval from sample indices using the formula
- ☐ I can calculate heart rate from RR interval
- ☐ I can explain WHY the R wave is used (not P or T)
- ☐ I can explain HRV and why it differs between Normal and AF
- ☐ I can name 4 types of ECG noise and their frequency ranges

- ☐ I can state the Nyquist theorem and apply it to ECG
- ☐ I understand the motion → false positive chain (our research!)
- ☐ I can explain mechanistically why AF causes strokes

▶ READY FOR PHASE 2: CACHET-CADB Dataset Study

Complete all 10 checklist items and answer 20/30 viva questions before proceeding.